

Code-Layout, Readability and Reusability

Date	01 july 2026
Team ID	SWTID-2026-2152
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Automated formatting applied via Black/autopep8 in VS Code.	Yes	Verified
2	Proper File Structure	Organized into /src, /data, /models, and /notebooks.	Yes	Standardized
3	Meaningful Variable Names	Tensors/Variables named: input_frame, predicted_emotion_label, confidence_score.	Yes	Descriptive
4	Function / Method Names	Action-oriented names: detect_emotion(), get_student_feedback(), save_model_weights().	Yes	Implemented
5	Code Comments	Documented the <i>why</i> (e.g., "# Normalizing pixel values to 0-1 range for faster convergence").	Yes	Thorough
6	Modular Design	Architecture separated: model.py (arch), capture.py (video logic), ui.py (interface).	Yes	Decoupled

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Face Detector	OpenCV / MTCNN	Frame processing	High
2	Emotion Model	TensorFlow / Keras	Emotion classification	High
3	Data Preprocessor	NumPy / PIL	Image transformation	High
4	UI Dashboard	Streamlit / Flask	Displaying feedback	Medium
5	Logger Utility	Python logging	System status/errors	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
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Code Layout & Structure	4/5	Modular structure separates UI, logic, and models effectively.
Readability	5/5	Consistent naming conventions used across all modules.
Reusability	4/5	Core components (detection/classification) are decoupled from the main app.
Documentation	3/5	Basic docstrings included; could benefit from more detailed README.
Overall Score	4/5	Solid, maintainable, and highly professional architecture.